PANTONE REPORT : THE PAGES OF MICHEL DAVID

PAGE 10

OPERATION OF AN EXPLOSION ENGINE WITH GAS OIL OR FUEL

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You saw on the photos on my page 8, that my little experimental burner worked perfectly on fuel oil with a pretty blue gas burner flame, without any smoke or smell, this as soon as the reactor is hot with the fuel sprayed in a simple gasoline carburetor.

As in this experiment I made it as simple as possible, that this small carburetor is not heated and is even very cooled by the spraying of the fuel, I have to put it in the choke position to ensure this spraying.

All this to tell you that I run my little Honda 240cc engine perfectly, which I use for all my experiments, with this same small carburetor and pure heating fuel.

As it is not possible to start cold, as the reactor is not at temperature, it is necessary to start the engine with gasoline to heat it for 2 or 3 minutes, and switch to fuel after this. time.

We can see that the engine behaves as well as gasoline and does not produce smoke or odor in the exhaust.

The engine therefore works well on gas resulting from the decomposition of fuel in the

reactor, as can be seen with the naked eye by the flame of my small burner with zero CO production and very little CO2.

Although legally you cannot run your gasoline car with gas oil or fuel transformed into gas by the PANTONE reactor, the business is still interesting for economically operating fixed-station engines such as small generators, tillers. and other equipment of all kinds, which are usually real fuel eaters and pollution champions.

For the engine equipment you want, you will have to take into account its displacement to define from the start the size of your reactor. Always build the smallest engine possible so that it warms up quickly and keeps up well at engine idle. Provide good thermal insulation of the exhaust outlet and the reactor with glass wool or other materials suitable for high temperatures.

You can keep the original carburetor of the engine which may work very well without modifying its choke position, but you will probably need, for everything to go well, modify the float setting so that the fuel level in the tank is as high as possible, obviously without overflow from the nozzle. You may have to put a smaller or larger nozzle, which you will determine by trial and error and by step. You will also have to grope on the position of the valve or butterfly, air inlet etc.

If you can get the carburetor to warm up a bit as well as the air flowing through it, that will be all the better.

As can be seen in the photos, when my engine works with a carburetor at the inlet of the reactor, it is equipped with my rotary valve system for entering the air to be mixed with the gas leaving the reactor, of which j explained the utility on my page 5.

Rotary air inlet valve

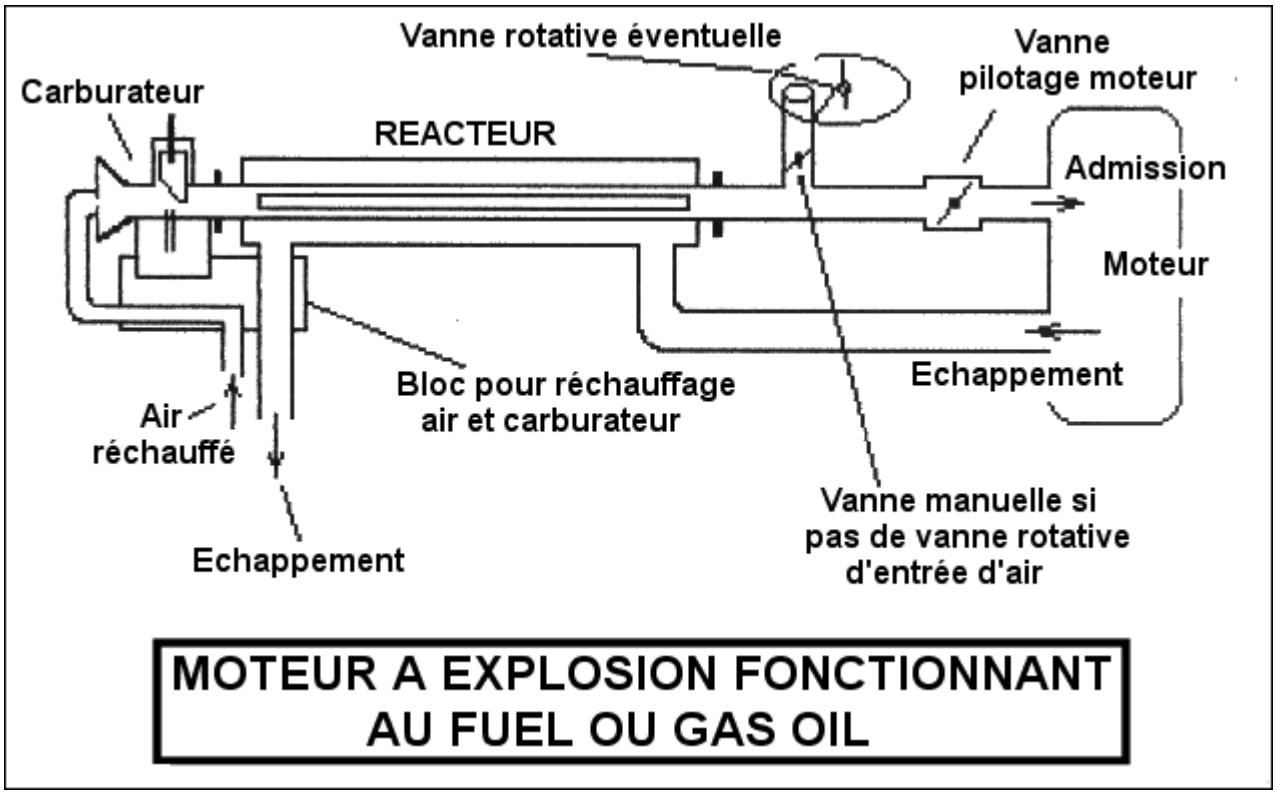
Understand that by briefly closing the fresh air intake, the suction through the reactor, and therefore into the carburetor, is increased in order to obtain the fuel atomization at low engine speed.

If you equip an engine with a somewhat high normal operating speed, such as a generator, you can do without this device.

You can obviously imagine something else to strengthen the suction in the reactor and therefore in the carburetor, which is in fact a rather difficult problem to solve in a simple and effective way, a solution that I am always looking for to offer it to you.

As I told you in my first pages, the invention of Mr. PANTONE, that is to say the reactor, does not pose any particular problem, it is all the peripheral equipment which is difficult to develop because of the multiple possible possibilities.

Now if you have the skills, you can replace the carburetor solution, accessible to all, by the injection solution, with pump, electronic pulse generator, sensors etc. It is this solution that I adopted for my boiler burner that I will describe to you in a next page.



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